Blockchain Consensus Benchmarking

Performance and Energy Efficiency Analysis

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Executive Summary

This report presents comprehensive benchmarking results for blockchain consensus workloads across multiple EC2 instance types. A total of 16 benchmarks were executed across 4 instance types and 6 workload types, measuring both performance (throughput) and energy efficiency (joules per work unit).

Best Performance: t3.2xlarge running pow achieved 572676.53 work units/second.
Most Energy Efficient: t4g.2xlarge running pow consumed only 0.000014 Joules per work unit.

Test Methodology

Overview

The benchmark suite tests four blockchain consensus algorithms across four EC2 instance types, measuring performance (throughput) and energy efficiency (energy per work unit).

Instance Types Tested

• **t4g.micro** (ARM): 2 vCPU, 1GB RAM - Burstable performance instance
• **t3.micro** (x86): 2 vCPU, 1GB RAM - Burstable performance instance
• **g4dn.xlarge** (GPU): 4 vCPU, 16GB RAM, NVIDIA T4 GPU
• **f1.2xlarge** (FPGA): 8 vCPU, 122GB RAM, Xilinx Virtex UltraScale+ VU9P

Workloads Tested

• **Proof of Work (PoW)**: SHA-256 mining with fixed difficulty target
• **Proof of Stake (PoS)**: Ed25519 signature verification
• **Byzantine Fault Tolerance (BFT)**: PBFT consensus simulation (4 nodes, 1 Byzantine tolerance)
• **Zero-Knowledge Proofs (ZK)**: ZK-SNARK proof generation simulation

Measurement Protocol

Each benchmark follows a standardized protocol to ensure comparability:
• **Warmup Period:** 30 seconds (excluded from results)
• **Measurement Period:** 5 minutes (300 seconds) - only this period counts
• **Fixed Parameters:** Same difficulty targets, key sizes, and workload configurations across all instances
• **Fresh Instances:** Each test uses a freshly provisioned instance to avoid caching effects
• **Real-time Monitoring:** CloudWatch metrics collected at 1-minute intervals

Performance Measurement

Performance is measured as **throughput**: the number of work units completed per second. Each workload counts distinct work units:
• PoW: SHA-256 hash attempts
• PoS: Ed25519 signature verifications
• BFT: Consensus rounds completed
• ZK: Proof generations completed

Energy Efficiency Measurement

Energy consumption is estimated using a power-based model:
• **Base Power:** AWS published TDP (Thermal Design Power) values
• **CPU Power:** Base TDP × (CPU Utilization / 100)
• **GPU Power:** Measured directly from nvidia-smi for GPU instances
• **Energy:** Power (Watts) × Time (seconds) = Energy (Joules)
• **Efficiency:** Energy (Joules) / Work Units = Joules per work unit
Note: Lower efficiency values indicate better energy efficiency.

Results Summary

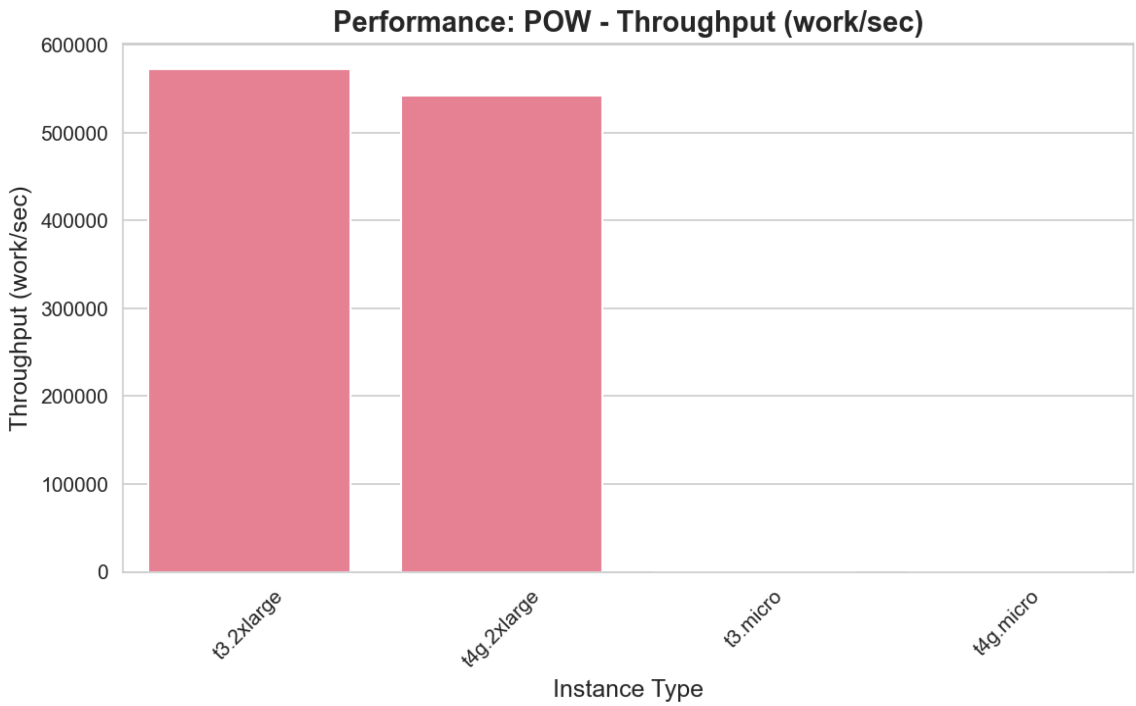
workload	work_units_completed	throughput_work_per_sec	efficiency_joules_per_work	cost_per_work_usd	average
pow	0	0.00e+00	N/A	N/A	0
zk	1802360	6007.8667	0.0014	1.54e-08	0
bft	52444791	174815.9700	4.12e-05	4.27e-10	0
pow	0	0.00e+00	N/A	N/A	0
pow	162809595	542698.6500	1.45e-05	1.38e-10	
bft	41441193	138137.3100	6.08e-05	6.69e-10	0
zk_cloudzk	834159	2780.5300	0.0026	2.69e-08	0
pow	171802959	572676.5300	1.47e-05	1.61e-10	
zk	1576602	5255.3400	0.0014	1.42e-08	0
pos	0	0.00e+00	N/A	N/A	0
pos	1824153	6080.5100	0.0012	1.23e-08	0
pos	0	0.00e+00	N/A	N/A	0
pow_gpu	0	0.00e+00	N/A	N/A	0
pow_gpu	0	0.00e+00	N/A	N/A	0
pos	1964782	6549.2733	0.0013	1.41e-08	0
zk_cloudzk	875489	2918.2967	0.0029	3.17e-08	0

Performance Analysis

POW Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t3.2xlarge	572676.5300	71584.5663	171802959	300
t4g.2xlarge	542698.6500	67837.3313	162809595	300
t3.micro	0.00e+00	0.00e+00	0	300
t4g.micro	0.00e+00	0.00e+00	0	300

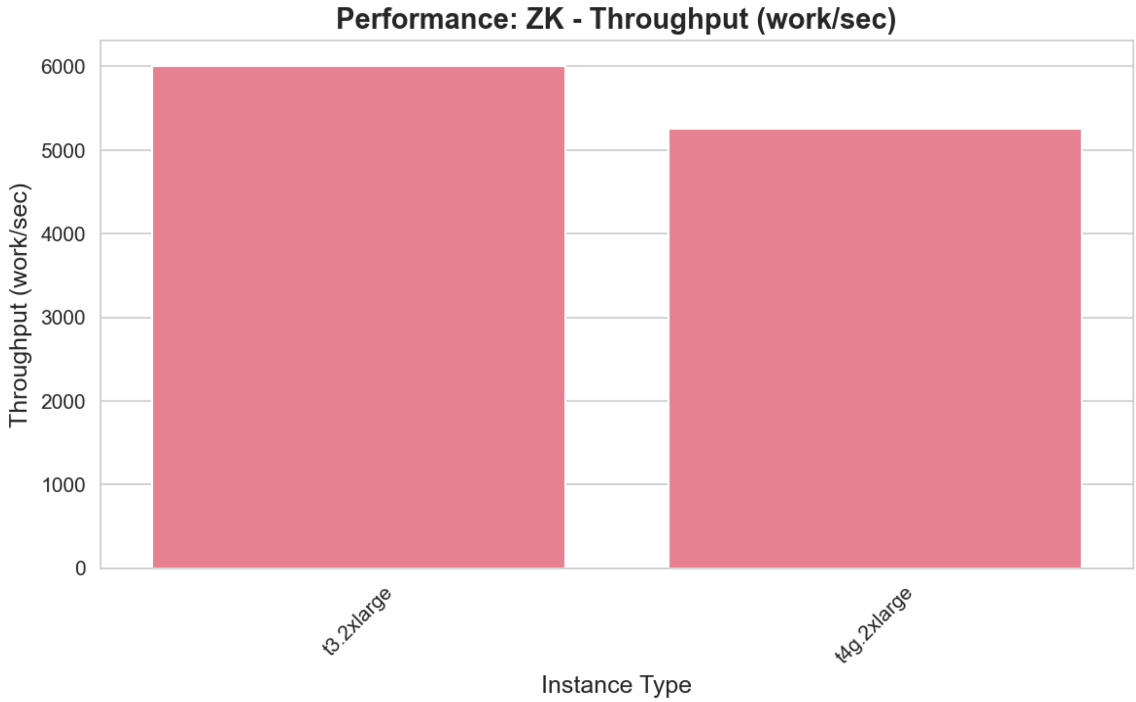
The **t3.2xlarge** instance achieved the highest throughput of 572676.53 work units/second for pow workloads, completing 171802959 work units in 300 seconds.



ZK Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t3.2xlarge	6007.8667	750.9833	1802360	300
t4g.2xlarge	5255.3400	656.9175	1576602	300

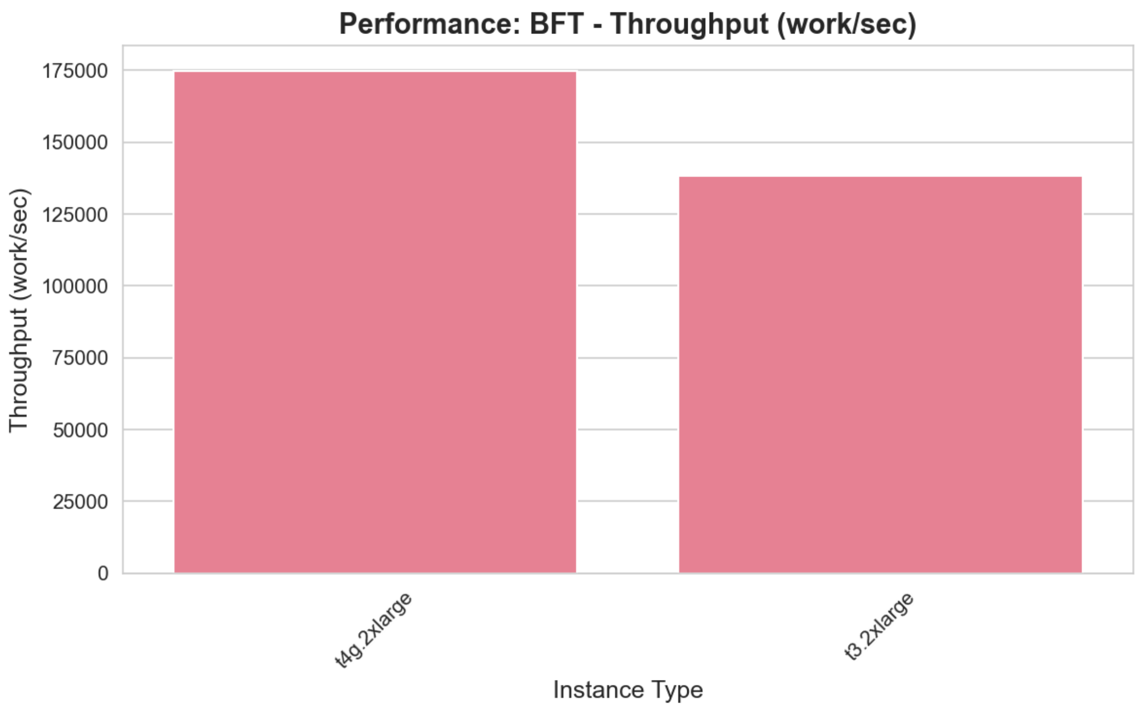
The **t3.2xlarge** instance achieved the highest throughput of 6007.87 work units/second for zk workloads, completing 1802360 work units in 300 seconds.



BFT Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t4g.2xlarge	174815.9700	21851.9963	52444791	300
t3.2xlarge	138137.3100	17267.1637	41441193	300

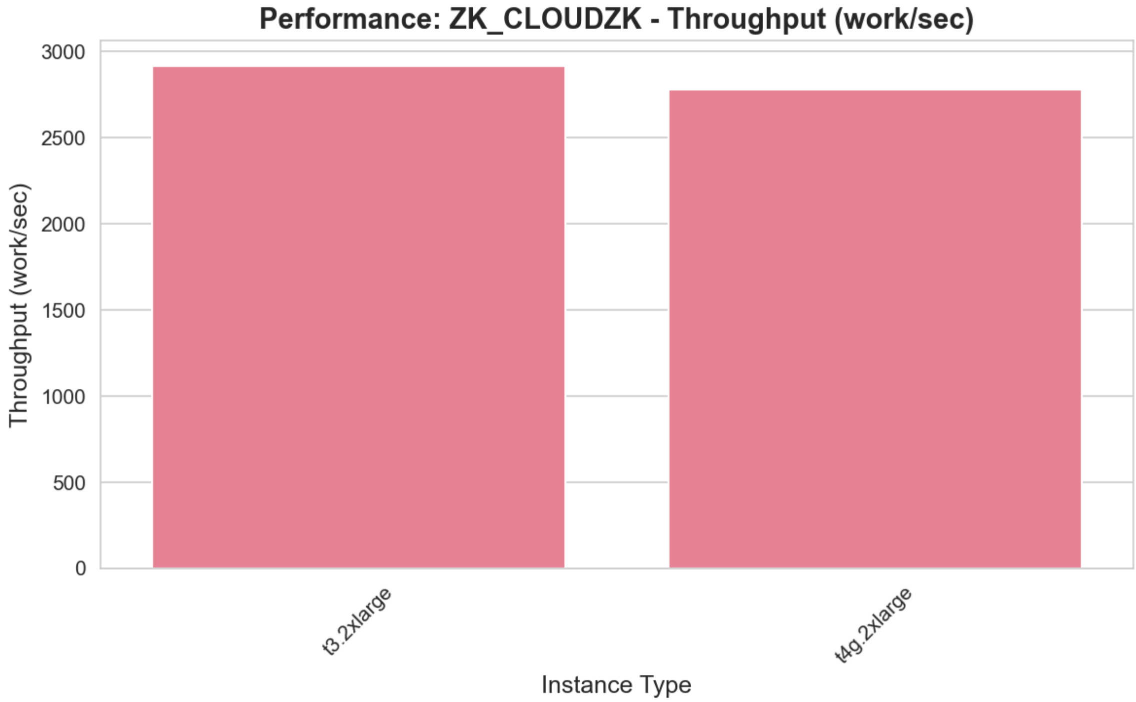
The **t4g.2xlarge** instance achieved the highest throughput of 174815.97 work units/second for bft workloads, completing 52444791 work units in 300 seconds.



ZK_CLOUDZK Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t3.2xlarge	2918.2967	364.7871	875489	300
t4g.2xlarge	2780.5300	347.5663	834159	300

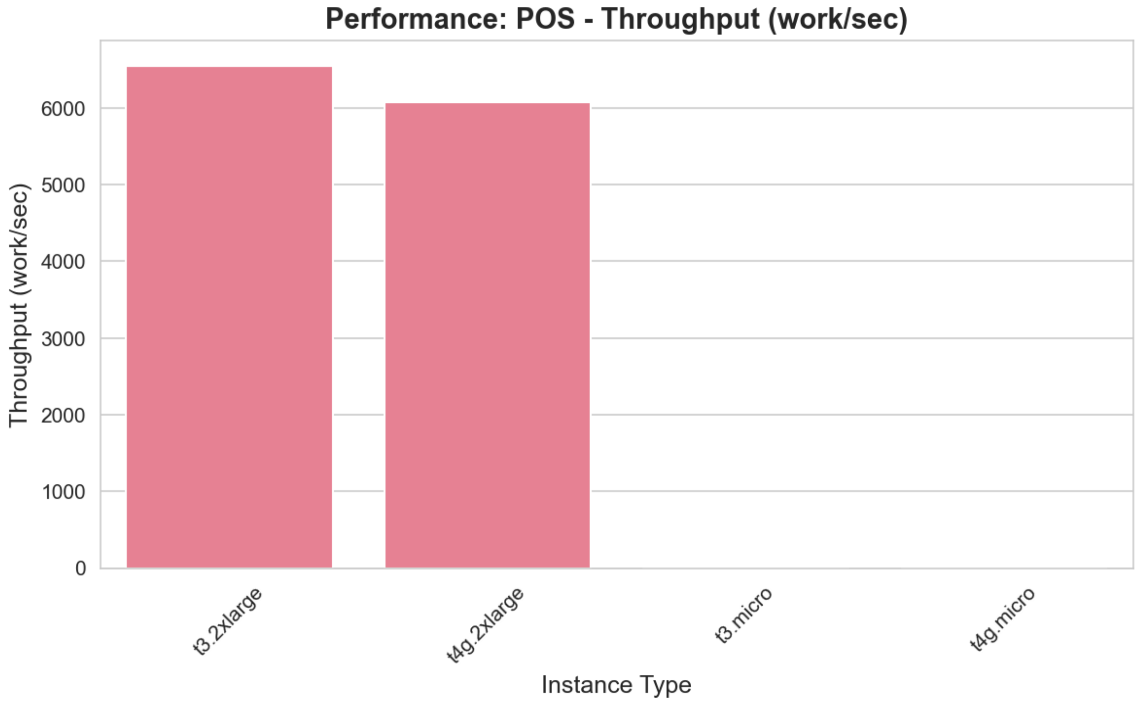
The **t3.2xlarge** instance achieved the highest throughput of 2918.30 work units/second for zk_cloudzk workloads, completing 875489 work units in 300 seconds.



POS Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t3.2xlarge	6549.2733	818.6592	1964782	300
t4g.2xlarge	6080.5100	760.0638	1824153	300
t3.micro	0.00e+00	0.00e+00	0	300
t4g.micro	0.00e+00	0.00e+00	0	300

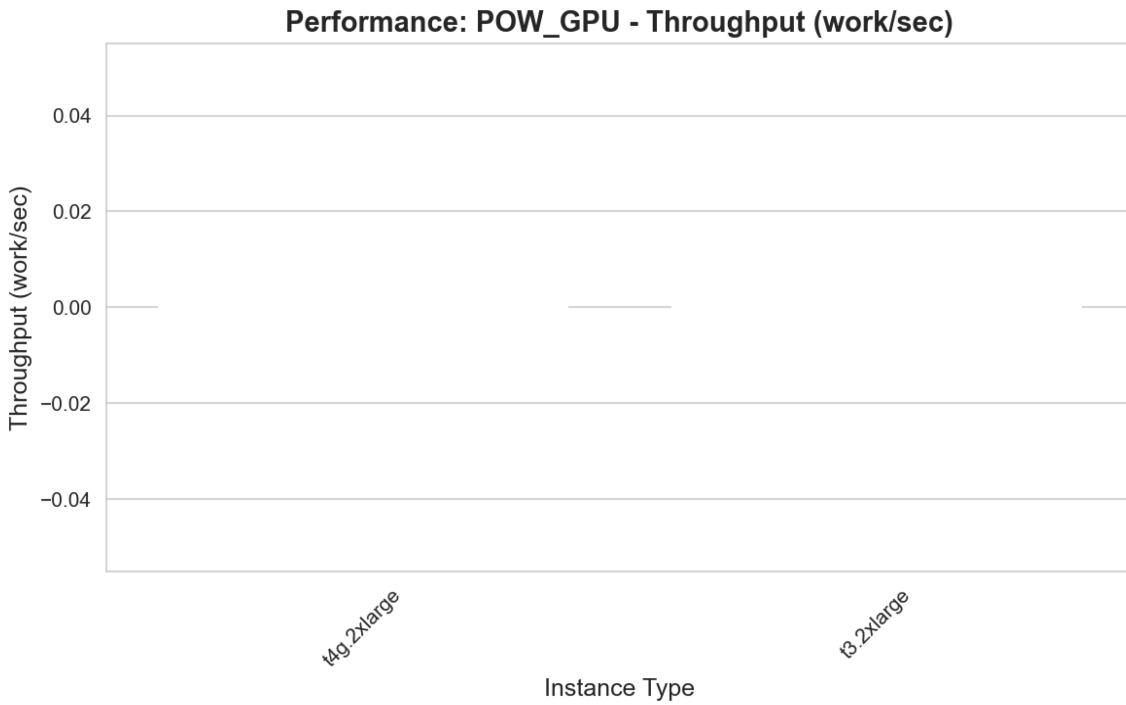
The **t3.2xlarge** instance achieved the highest throughput of 6549.27 work units/second for pos workloads, completing 1964782 work units in 300 seconds.



POW_GPU Workload

instance_type	throughput_work_per_sec	throughput_per_vcpu	work_units_completed	measurement_duration_seconds
t4g.2xlarge	0.00e+00	0.00e+00	0	300
t3.2xlarge	0.00e+00	0.00e+00	0	300

The **t4g.2xlarge** instance achieved the highest throughput of 0.00 work units/second for pow_gpu workloads, completing 0 work units in 300 seconds.

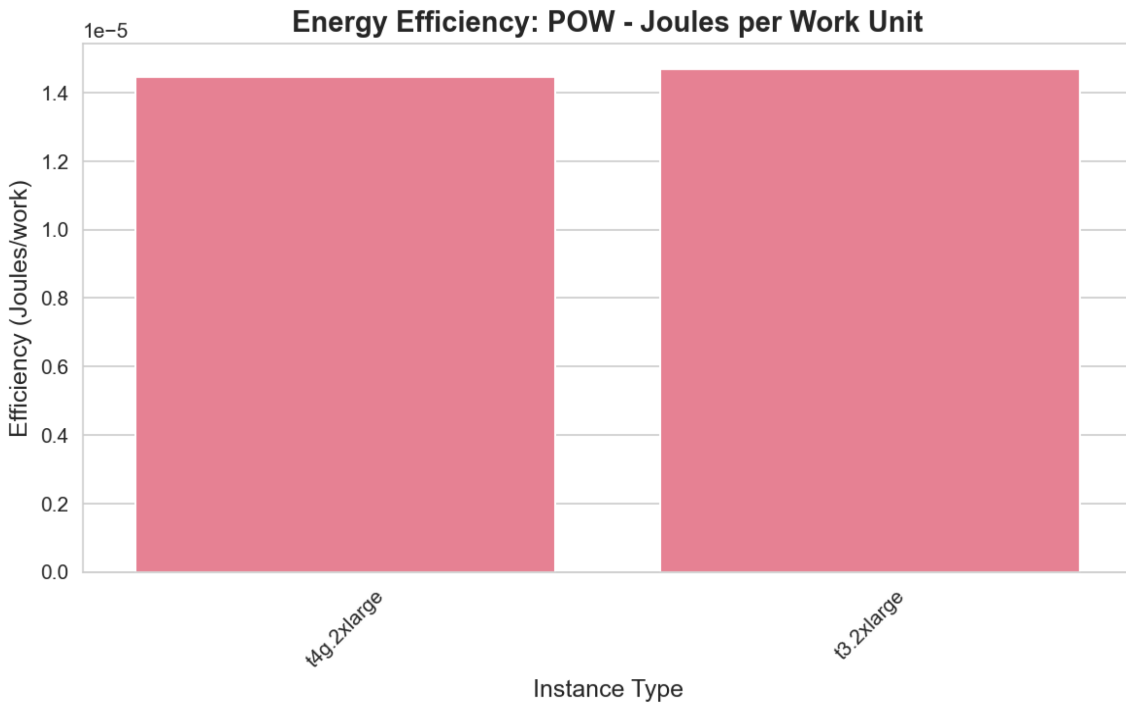

Performance: POW_GPU - Throughput (work/sec)
0.04
0.02
0.00
−0.02
−0.04
Throughput (work/sec)
t4g.2xlarge
t3.2xlarge
Instance Type

Energy Efficiency Analysis

POW Workload

instance_type	efficiency_joules_per_work	energy_joules	work_units_completed	average_cpu_percent
t4g.2xlarge	1.45e-05	2353.7170	162809595	3.8436
t3.2xlarge	1.47e-05	2524.6013	171802959	0.0783

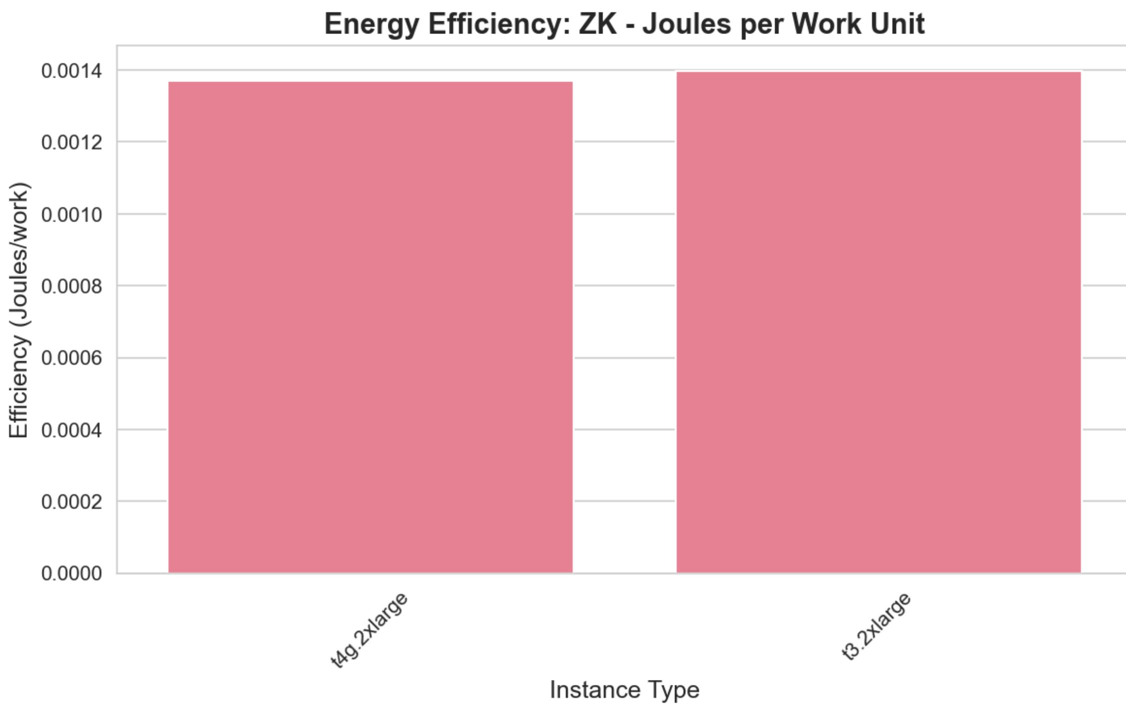
The **t4g.2xlarge** instance was the most energy-efficient, consuming only 0.000014 Joules per work unit. Total energy consumption was 2353.72 Joules for 162809595 work units.



ZK Workload

instance_type	efficiency_joules_per_work	energy_joules	work_units_completed	average_cpu_percent
t4g.2xlarge	0.0014	2160.0000	1576602	0.00e+00
t3.2xlarge	0.0014	2520.0000	1802360	0.00e+00

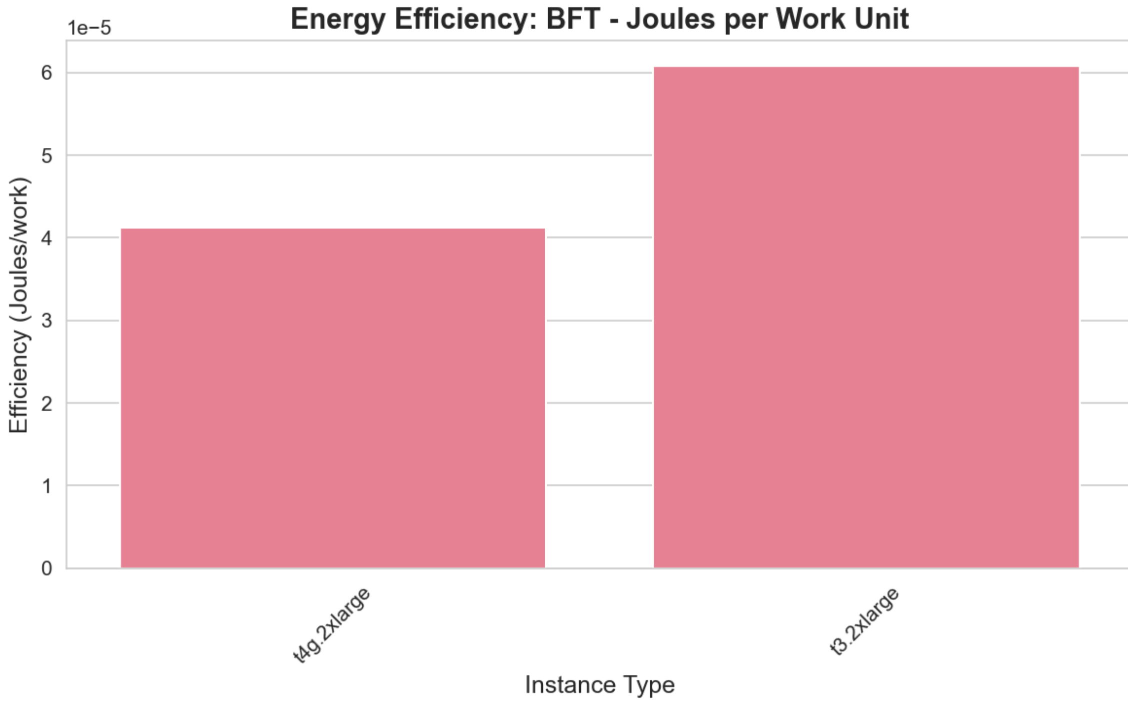
The **t4g.2xlarge** instance was the most energy-efficient, consuming only 0.001370 Joules per work unit. Total energy consumption was 2160.00 Joules for 1576602 work units.



BFT Workload

instance_type	efficiency_joules_per_work	energy_joules	work_units_completed	average_cpu_percent
t4g.2xlarge	4.12e-05	2160.0000	52444791	0.00e+00
t3.2xlarge	6.08e-05	2520.0000	41441193	0.00e+00

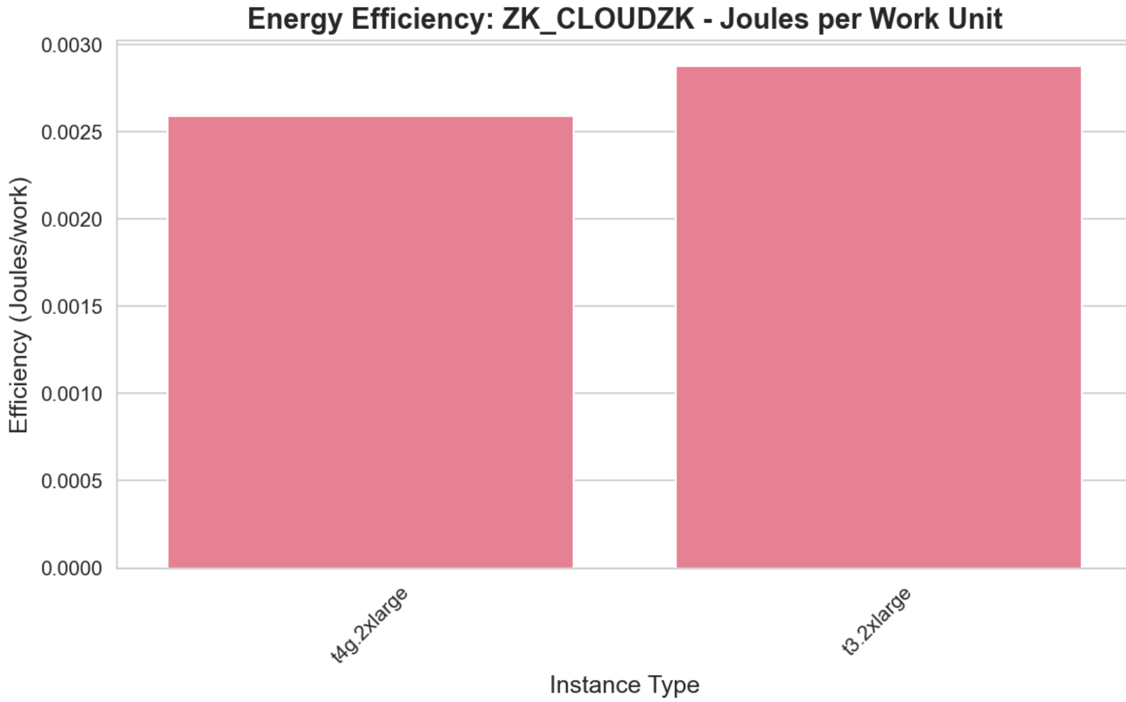
The **t4g.2xlarge** instance was the most energy-efficient, consuming only 0.000041 Joules per work unit. Total energy consumption was 2160.00 Joules for 52444791 work units.



ZK_CLOUDZK Workload

instance_type	efficiency_joules_per_work	energy_joules	work_units_completed	average_cpu_percent
t4g.2xlarge	0.0026	2160.0000	834159	0.00e+00
t3.2xlarge	0.0029	2520.0000	875489	0.00e+00

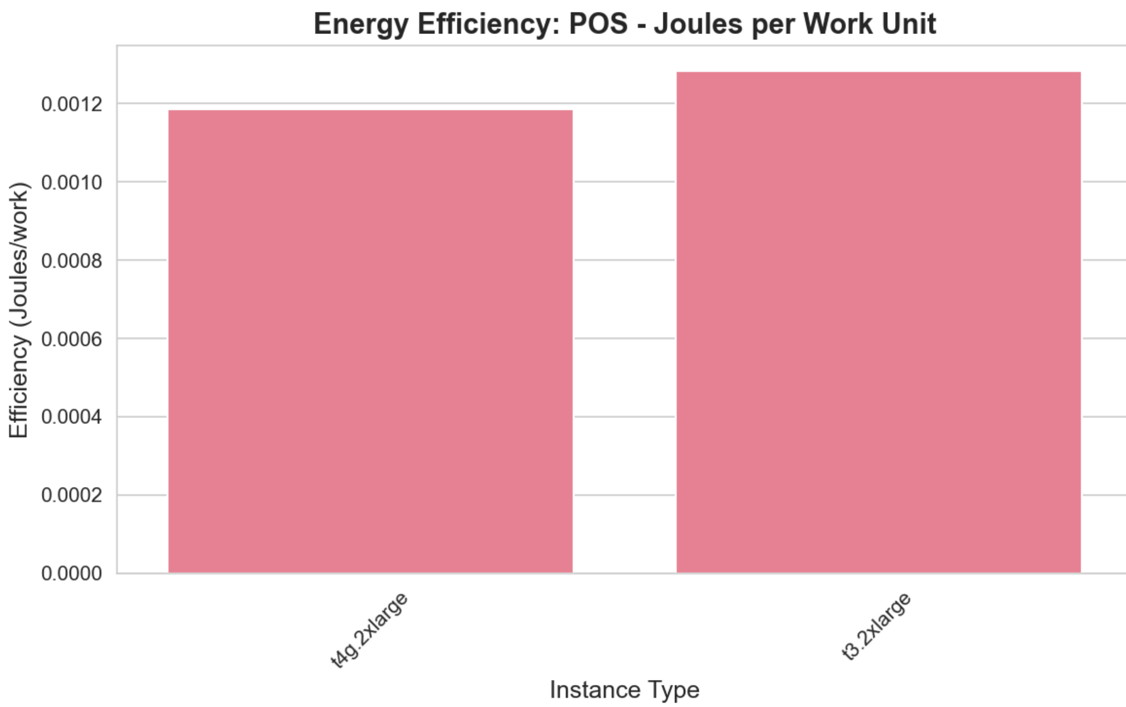
The **t4g.2xlarge** instance was the most energy-efficient, consuming only 0.002589 Joules per work unit. Total energy consumption was 2160.00 Joules for 834159 work units.



POS Workload

instance_type	efficiency_joules_per_work	energy_joules	work_units_completed	average_cpu_percent
t4g.2xlarge	0.0012	2160.0000	1824153	0.00e+00
t3.2xlarge	0.0013	2520.0000	1964782	0.00e+00

The **t4g.2xlarge** instance was the most energy-efficient, consuming only 0.001184 Joules per work unit. Total energy consumption was 2160.00 Joules for 1824153 work units.



POW_GPU Workload

Detailed Results and Verbose Logs

This section contains detailed logs from 16 benchmark runs.

Benchmark 1: t3.micro / pow

Instance ID: i-04bbaa19d84b0e26a
Public IP: 18.206.92.8
Measurement Start: 2025-11-20T18:06:50.899449
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: N/A%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 2: t3.2xlarge / zk

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:42:04.466388
Duration: 300 seconds

Workload Output:
• Work units completed: 1802360
• Throughput: 6007.87 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 3: t4g.2xlarge / bft

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:36:33.012800
Duration: 300 seconds

Workload Output:
• Work units completed: 52444791
• Throughput: 174815.97 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.7%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 4: t4g.micro / pow

Instance ID: i-0175cc718c01f1a54
Public IP: 35.173.129.40
Measurement Start: 2025-11-20T18:06:50.490438
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: N/A%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 5: t4g.2xlarge / pow

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:25:27.384744
Duration: 300 seconds

Workload Output:
• Work units completed: 162809595
• Throughput: 542698.65 work/sec
• Energy consumed: 276.74 Joules
• Efficiency: 0.000002 Joules/work

Resource Metrics:
• Average CPU: 3.8%
• Max CPU: 3.8%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 6: t3.2xlarge / bft

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:36:32.615516
Duration: 300 seconds

Workload Output:
• Work units completed: 41441193
• Throughput: 138137.31 work/sec

• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 7: t4g.2xlarge / zk_cloudzk

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:47:36.753107
Duration: 300 seconds

Workload Output:
• Work units completed: 834159
• Throughput: 2780.53 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 8: t3.2xlarge / pow

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:25:27.907423
Duration: 300 seconds

Workload Output:
• Work units completed: 171802959
• Throughput: 572676.53 work/sec
• Energy consumed: 6.57 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: 0.1%
• Max CPU: 0.1%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 9: t4g.2xlarge / zk

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:42:04.829257
Duration: 300 seconds

Workload Output:
• Work units completed: 1576602
• Throughput: 5255.34 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 10: t3.micro / pos

Instance ID: i-0c89838a945ffe028
Public IP: 44.197.203.17
Measurement Start: 2025-11-20T17:53:01.607169
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 18.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 11: t4g.2xlarge / pos

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:31:00.860976
Duration: 300 seconds

Workload Output:
• Work units completed: 1824153
• Throughput: 6080.51 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 12: t4g.micro / pos

Instance ID: i-03f0803358cfa54d0

Public IP: 54.226.218.180
Measurement Start: 2025-11-20T17:53:01.519439
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 14.7%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 13: t4g.2xlarge / pow_gpu

Instance ID: i-0555d26cd56a6ca02
Public IP: 34.224.26.127
Measurement Start: 2025-11-20T20:53:08.547910
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 14: t3.2xlarge / pow_gpu

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:53:08.446659
Duration: 300 seconds

Workload Output:
• Work units completed: N/A
• Throughput: 0.00 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: N/A Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 15: t3.2xlarge / pos

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:31:00.439403
Duration: 300 seconds

Workload Output:
• Work units completed: 1964782
• Throughput: 6549.27 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Benchmark 16: t3.2xlarge / zk_cloudzk

Instance ID: i-005c67216c4dd87d3
Public IP: 44.222.171.117
Measurement Start: 2025-11-20T20:47:36.622040
Duration: 300 seconds

Workload Output:
• Work units completed: 875489
• Throughput: 2918.30 work/sec
• Energy consumed: 0.00 Joules
• Efficiency: 0.000000 Joules/work

Resource Metrics:
• Average CPU: N/A%
• Max CPU: N/A%
• Memory Usage: 0.6%
• GPU Utilization: N/A%
• GPU Power: N/AW

Conclusions and Recommendations

Key Findings

POW Workload:
For maximum performance, use **t3.2xlarge** (572676.53 work/sec). For optimal energy efficiency, use **t4g.2xlarge** (0.000014 J/work).

ZK Workload:
For maximum performance, use **t3.2xlarge** (6007.87 work/sec). For optimal energy efficiency, use **t4g.2xlarge** (0.001370 J/work).

BFT Workload:
t4g.2xlarge provides both the best performance (174815.97 work/sec) and energy efficiency (0.000041 J/work if available) for bft workloads.

ZK_CLOUDZK Workload:
For maximum performance, use **t3.2xlarge** (2918.30 work/sec). For optimal energy efficiency, use **t4g.2xlarge** (0.002589 J/work).

POS Workload:
For maximum performance, use **t3.2xlarge** (6549.27 work/sec). For optimal energy efficiency, use **t4g.2xlarge** (0.001184 J/work).

POW_GPU Workload:
t4g.2xlarge provides both the best performance (0.00 work/sec) and energy efficiency (N/A J/work if available) for pow_gpu workloads.

Trade-offs

Higher-performance instances typically consume more energy. The choice between instance types depends on priorities:

- **Performance-critical:** Choose instances with highest throughput
- **Energy-constrained:** Choose instances with lowest Joules/work
- **Cost-optimized:** Balance performance and cost per work unit
- **Balanced:** Consider throughput per vCPU and efficiency per vCPU for normalized comparison

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